Artificial Intelligence & Machine Learning

Rising Waters: A Machine Learning Approach to Flood Prediction

1. Introduction

- **Project Title:** Rising Waters – A Machine Learning Approach to Flood Prediction
- **Team Members:**
  - Ramya Paila
  - Tanniru Gnana Sindhuja
  - Valluri Harini
  - Ravuri Hiranmai

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2. Project Overview

Purpose

Rising Waters is an intelligent flood prediction system that analyzes environmental and rainfall parameters to determine the likelihood of severe flooding. It enables early warnings and supports disaster preparedness.

Key Features

- Real-time flood risk prediction
- Machine learning classification model
- Web interface for user input
- Instant prediction results
- Lightweight and fast execution

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3. System Architecture

Frontend

- HTML5 for structure
- CSS3 for styling
- Responsive UI with card-based layout
- Form inputs for environmental parameters

Backend

- Flask web framework
- Python server-side logic
- Routes for navigation and prediction

Machine Learning Layer

- Model trained using classification algorithms:
  - Decision Tree
  - Random Forest
  - KNN
  - XGBoost
- Best performing model selected and saved using Pickle/Joblib

Data Flow

User Input → Flask Server → Data Scaling → ML Model → Prediction → Result Page

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4. Technologies Used

- Python
- Flask
- NumPy
- Scikit-learn
- XGBoost
- Joblib / Pickle
- HTML
- CSS

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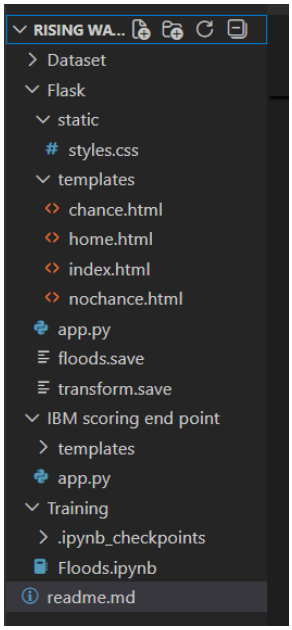
5. Project Folder Structure



6. Running the Application

- python app.py

Open browser:

- http://127.0.0.1:5000

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7. Input Parameters

The system requires environmental data:

- Temperature
- Humidity
- Cloud Cover
- Annual Rainfall
- Jan–Feb Rainfall
- Mar–May Rainfall
- Jun–Sep Rainfall
- Oct–Dec Rainfall

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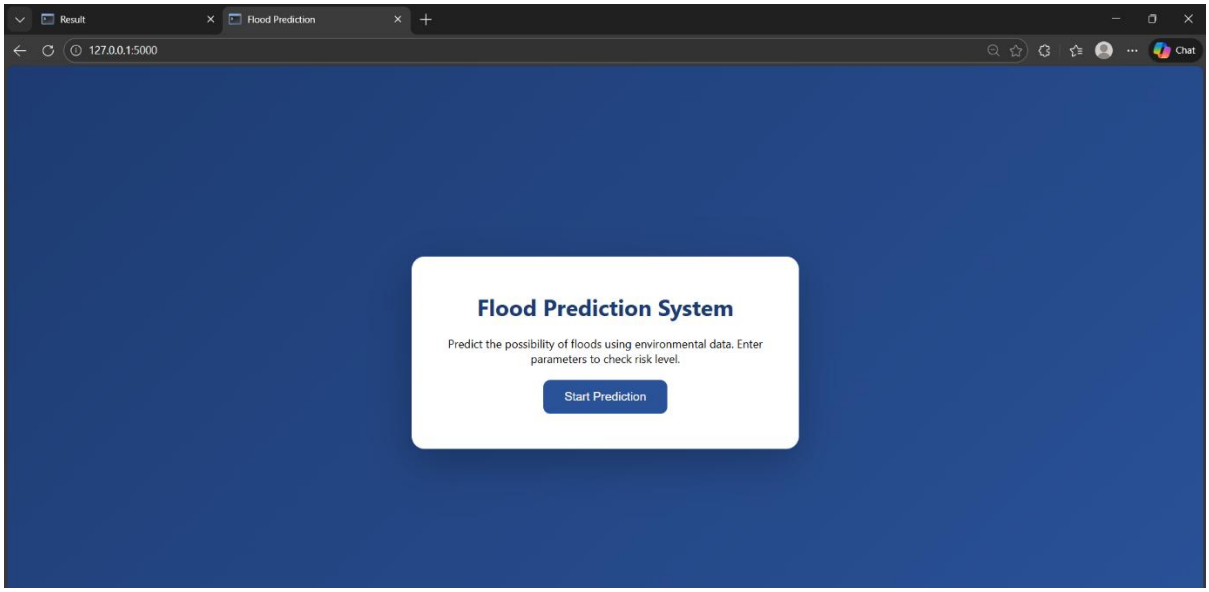
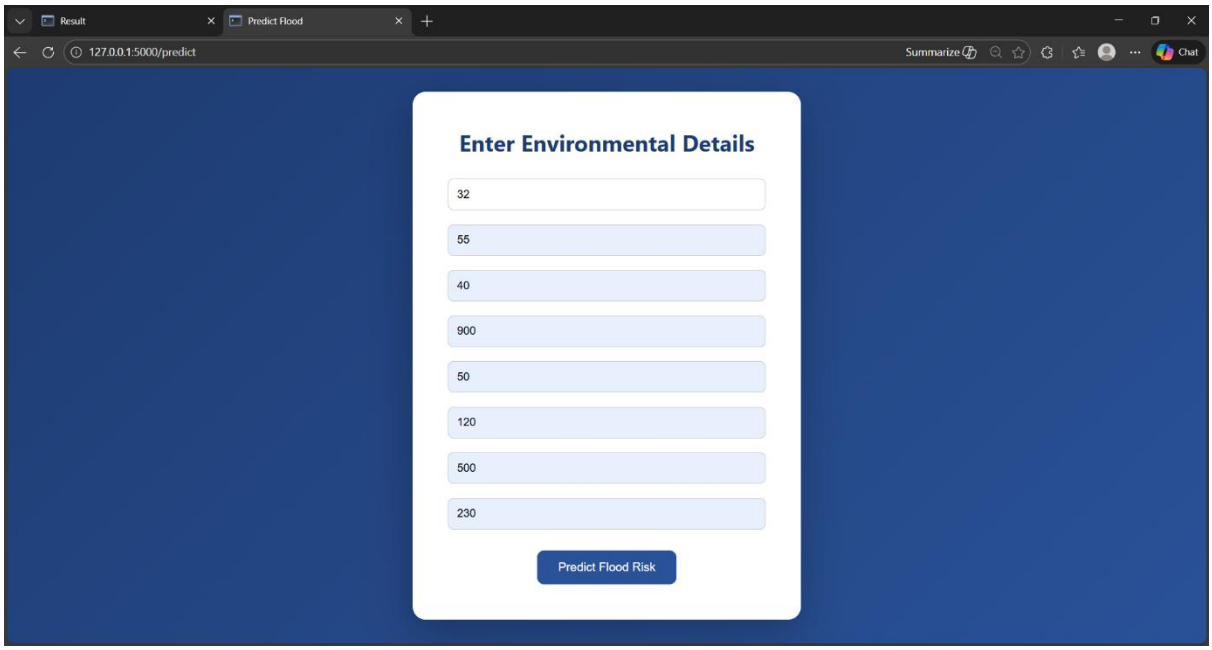
8. Output & UI

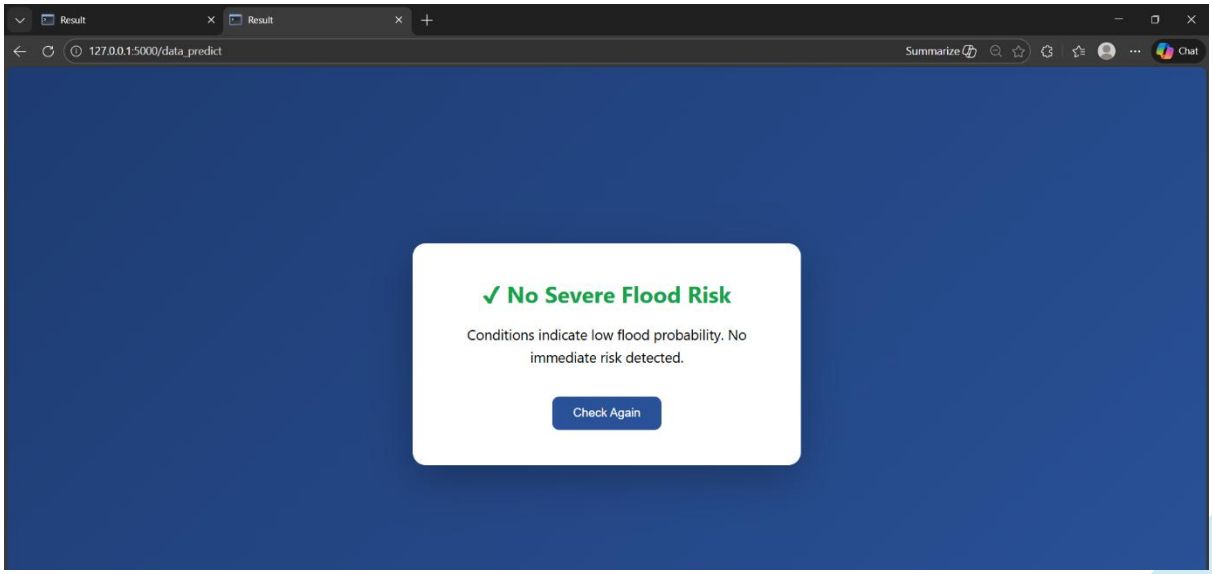
The model predicts one of two results:

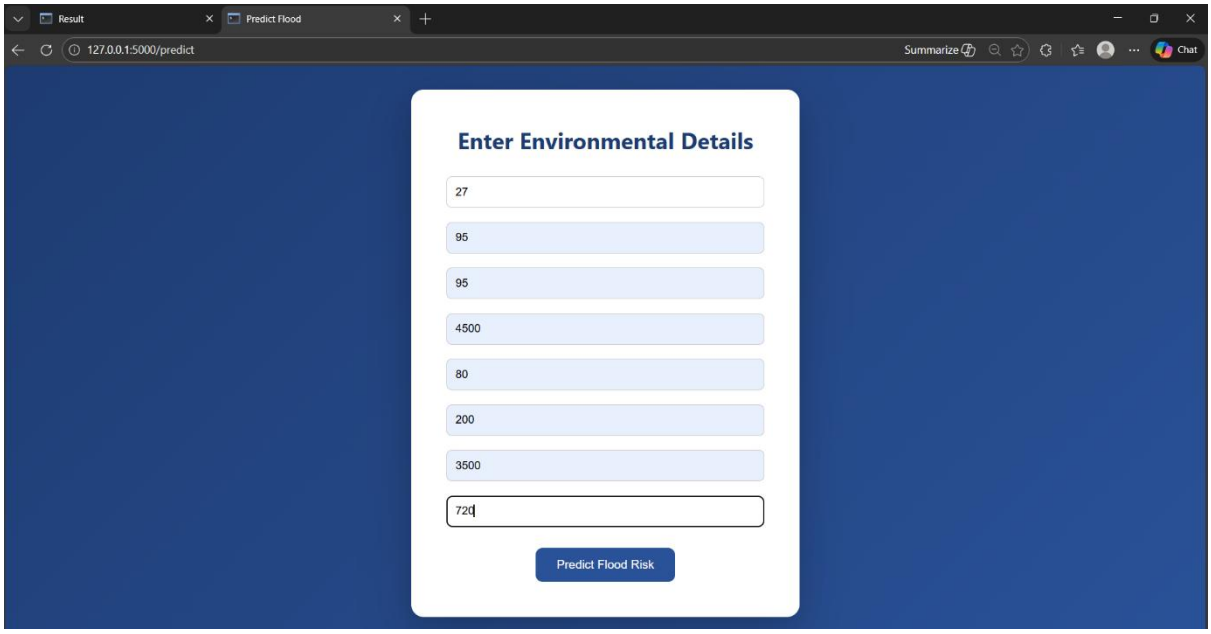
- **Severe Flood Risk** → High probability of flood
- **No Severe Flood Risk** → Low probability

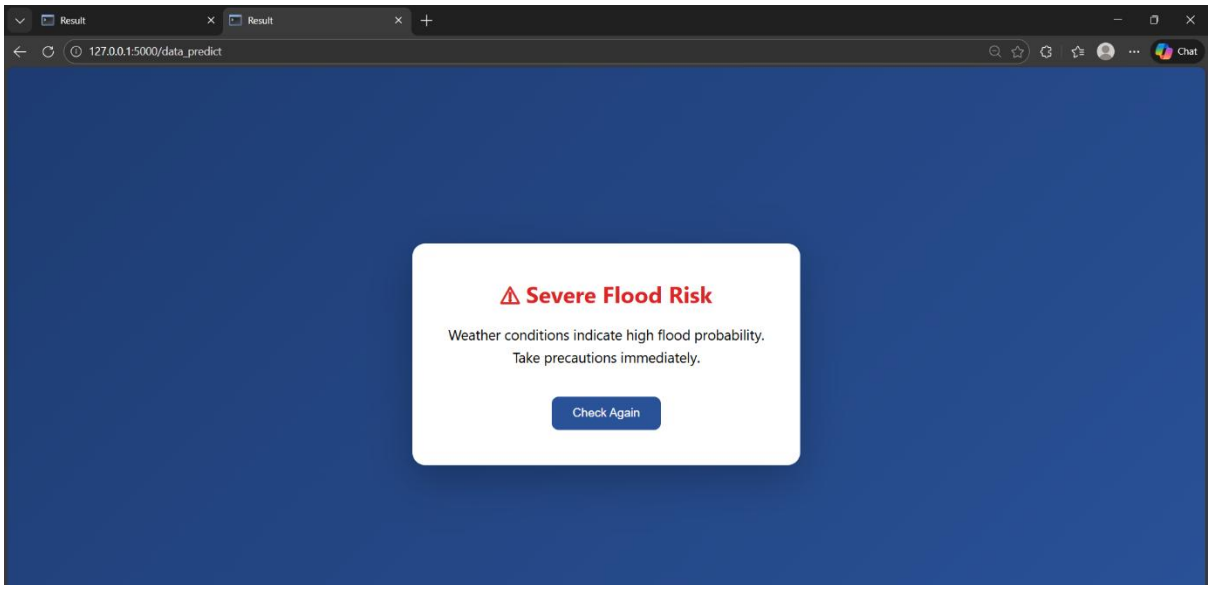
```
 * Serving Flask app "app" (lazy loading)
 * Environment: production
   WARNING: This is a development server. Do not use it in a production deployment.
   Use a production WSGI server instead.
 * Debug mode: on
 * Restarting with stat
C:\Users\hiran\AppData\Local\Programs\Python\Python311\Lib\site-packages\sklearn\base.py:463: InconsistentVersionWarning: Trying to unpickle estimator StandardScaler fr
om version 1.6.1 when using version 1.8.0. This might lead to breaking code or invalid results. Use at your own risk. For more info please refer to:
https://scikit-learn.org/stable/model_persistence.html#security-maintainability-limitations
  warnings.warn(
 * Debugger is active!
 * Debugger PIN: 266-096-065
 * Running on http://127.0.0.1:5000/ (Press CTRL+C to quit)
```


Result
Flood Prediction
127.0.0.1:5000
Chat
Flood Prediction System
Predict the possibility of floods using environmental data. Enter parameters to check risk level.
Start Prediction


Result
Predict Flood
127.0.0.1:5000/predict
Summarize
Chat
Enter Environmental Details
32
55
40
900
50
120
500
230
Predict Flood Risk


Result
Result
127.0.0.1:5000/data_predict
Summarize
Chat
✓ No Severe Flood Risk
Conditions indicate low flood probability. No immediate risk detected.
Check Again





9. Machine Learning Workflow

1. Data Collection
2. Data Cleaning
3. Exploratory Data Analysis
4. Feature Selection
5. Train/Test Split
6. Model Training
7. Model Evaluation
8. Best Model Saving
9. Deployment with Flask

10. Evaluation Metrics Used

- Accuracy Score
- Precision Score
- Recall Score
- Confusion Matrix

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11. Use Case Scenarios

- **Early Warning Systems**
  Authorities can issue alerts before floods occur.
- **Disaster Response Planning**
  Emergency teams can allocate resources effectively.
- **Infrastructure Planning**
  Engineers can design flood-resistant systems.

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12. Advantages

- Fast predictions
- Easy-to-use interface
- Useful for disaster management
- Scalable architecture

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13. Limitations

- Depends on data quality
- Requires retraining for new regions
- Prediction accuracy varies with input reliability

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14. Future Enhancements

- Live weather API integration
- Interactive dashboards
- Mobile application
- Cloud deployment

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15. Conclusion

Rising Waters demonstrates how machine learning can be integrated with web technologies to build a real-world disaster prediction system. It highlights the practical application of AI in public safety and environmental monitoring.